SOUND DECISION PROCEDURES FOR EQUATIONAL IMPLICATION OVER MAGMAS: KNUTH-BENDIX COMPLETION AND SMT MODEL FINDING

ANDREW REN AND NEURICO

Abstract. We study the problem of classifying equational implication over magmas: given two equational laws E and E' over a single binary operation, determine whether every magma satisfying E also satisfies E'. This question arises at scale in the Equational Theories Project, which has verified all 22,028,942 pairwise implications among 4,694 laws of order at most four.

We present predictor4, a layered decision procedure that replaces two unsound or incomplete components of its predecessor with theoretically grounded alternatives. For the positive implication path, Knuth-Bendix (KB) completion replaces a bounded breadth-first reachability search, providing a proof procedure that is *sound by construction* when it terminates. For the negative path, an SMT-based finite model finder using Z3 replaces a heuristic hill-climbing search, providing *completeness* for each fixed domain size.

A central contribution is the identification and formal analysis of a *soundness invariant* for KB completion: every rewrite rule $\ell \to r$ in the generated term rewriting system must satisfy $\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$. We prove that violation of this invariant causes all terms to collapse to the same normal form, making the normal-form equality test trivially true and producing false positives. We state this as a precise theorem and provide its proof.

Empirical evaluation on stratified samples of 50 and 200 cases from the ground-truth implication matrix confirms that predictor4, after enforcing the invariant, achieves 100% accuracy with zero false positives and zero false negatives, matching its predecessor on both sample sizes while providing strictly stronger theoretical guarantees for the positive implication path.

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1. Introduction

Universal algebra studies classes of algebraic structures defined by equational laws: identities of the form $s = t$ where s and t are terms built from variables and operation symbols. A fundamental question is the *implication problem*: given equational laws E and E', does E logically entail E'? That is, does every model of E also satisfy E'?

The *Equational Theories Project* (ETP) [4] recently determined the complete implication relation among 4,694 equational laws over a single binary operation $*$ (a *magma*) of *order at most four* (at most four applications of $*$), verifying each of the 22,028,942 pairwise implication decisions with formal Lean 4 proofs. This dataset provides a unique benchmark: a complete, machine-verified ground truth against which any automated implication classifier can be tested.

The classification problem. A practical implication classifier must be both *sound* (it never asserts a false implication) and *complete* (it never misses a true implication or counterexample). Full decidability of the implication problem for unrestricted magma laws is known to be unattainable [9]; nonetheless, efficient heuristic and semi-decision procedures are valuable for interactive algebraic exploration and automated hypothesis testing.

Soundness gaps in prior classifiers. The predecessor classifier PREDICTOR3 employs two components that are either unsound or incomplete. First, a bounded breadth-first reachability search serves as its positive implication test. While sound in principle—a reachable term intersection genuinely witnesses an equational derivation—its bounded character means it can spuriously report an implication when deep rewrite chains cause the reachable-term sets to collide accidentally, without a true proof. Second, a heuristic hill-climbing search tests for counterexamples in magmas of sizes 4 and 5; because hill-climbing is not exhaustive, it may terminate without finding a genuine counterexample of those sizes.

Our approach. We replace both unsound or incomplete components with established sound and complete procedures from the term rewriting and satisfiability literatures.

For the *positive path*, we implement Knuth-Bendix (KB) completion [8] from scratch in Python. When KB completion terminates with a convergent term rewriting system R, the normal-form equality test $\mathrm{NF}_R(\ell_2) = \mathrm{NF}_R(r_2)$ is a sound proof of $E \models E'$ by the Knuth-Bendix soundness theorem and Birkhoff's completeness theorem [3]. The bounded BFS search serves as a fallback for the approximately 88% of cases where KB does not terminate within the allotted time.

For the *negative path*, we encode the finite model finding problem as an SMT query in Z3 [5] for domain sizes $n = 4$ and $n = 5$. Z3 is complete for each fixed finite domain: if no counterexample of size n exists, Z3 returns UNSAT, providing stronger evidence for implication than a heuristic search that merely fails to find a witness.

The soundness invariant. During evaluation, we discovered that a buggy KB implementation generated rewrite rules $\ell \to r$ with $\mathrm{vars}(r) \not\subseteq \mathrm{vars}(\ell)$—specifically, rules of the form $x \to y$ where x and y are distinct variables. Such rules are not valid term rewriting rules under any reduction order, yet they can arise from incorrect variable renaming in the critical pair computation. We prove (Theorem 3.1) that any TRS containing a rule with a free RHS variable collapses all normal forms to a single term, making the normal-form test trivially true for all equation pairs and producing an unlimited supply of false positives. Enforcing the invariant $\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$ before accepting any rule eliminates this failure mode.

Main contributions.

- We formalize the *soundness invariant* for KB completion in the equational implication setting and prove that its violation causes total normal-form collapse (Theorem 3.1).
- We prove that, under the soundness invariant, KB completion provides a genuine proof of equational implication with no false positives (Theorem 3.5).
- We establish Z3's completeness for finite magma counterexample search at fixed domain sizes (Proposition 2.19), replacing heuristic hill-climbing with a decision procedure.
- We show that the resulting layered classifier PREDICTOR4 produces no false positives from its KB component and achieves 100% accuracy on stratified samples of 50 and 200 implications from the ETP ground truth (Theorem 4.1 and Section 5).

Organization. Section 2 establishes notation and the necessary background on magmas, equational implication, and term rewriting. Section 3 presents the main theoretical results. Section 4 describes the layered decision procedure PREDICTOR4. Section 5 reports the empirical evaluation. Section 6 discusses limitations and open questions, and Section 7 concludes.

2. Preliminaries

We fix notation and recall standard definitions from universal algebra and term rewriting theory. The reader may consult Baader and Nipkow [2] for a comprehensive treatment.

2.1. Magmas and Equational Laws.

Definition 2.1. A *magma* is a pair $(M, *)$ where M is a nonempty set and $*\colon M \times M \to M$ is a binary operation. We impose no additional axioms.

Let $\mathcal{X} = \{x, y, z, w, \ldots\}$ be a countably infinite set of *variables* and let $\Sigma = \{*\}$ be the signature consisting of a single binary operation symbol. The set $T(\Sigma, \mathcal{X})$ of *terms* over Σ and $\mathcal{X}$ is defined inductively: every variable $x \in \mathcal{X}$ is a term, and if s, t are terms then $s * t$ is a term.

Definition 2.2. An *equational law* is a pair of terms $(s, t) \in T(\Sigma, \mathcal{X})^2$, written $s = t$. An equational law $s = t$ is *trivial* if s and t are identical as terms.

Definition 2.3. A magma $(M, *)$ *satisfies* an equational law $s = t$, written $M \models (s = t)$, if for every assignment $\varphi\colon \mathcal{X} \to M$ of variables to elements, the evaluations $\varphi(s)$ and $\varphi(t)$ in M coincide. Here φ extends to all terms by $\varphi(u * v) = \varphi(u) * \varphi(v)$.

Definition 2.4. An equational law E *implies* an equational law E', written $E \models E'$, if every magma satisfying E also satisfies E'. We write $E \not\models E'$ for the negation; in this case any magma satisfying E but not E' is a *counterexample* to the implication.

Example 2.5. Let E be the *left-zero* law $x * y = x$ and let E' be $x * y = y * x$ (commutativity). Then $E \models E'$: if $a * b = a$ for all a, b then $a * b = a = b * a$ for all a, b. Conversely, let E be commutativity. The magma on $\{0, 1\}$ with $0*0 = 1, 0*1 = 1, 1 * 0 = 1, 1 * 1 = 0$ satisfies commutativity but not left-zero, so commutativity does not imply left-zero.

Remark 2.6. The implication relation $\models$ is a preorder on equational laws. The trivial law $x = x$ is satisfied by every magma and hence is implied by every law. The law $x = y$ (identifying all elements) is satisfied only by singleton magmas and implies every other law.

2.2. **Term Rewriting Systems.**

Definition 2.7. A *rewrite rule* is an ordered pair (ℓ, r) of terms written $\ell \to r$. A *term rewriting system* (TRS) is a finite set R of rewrite rules.

For a rule $\ell \to r$, let $\mathrm{vars}(\ell)$ and $\mathrm{vars}(r)$ denote the sets of variables occurring in ℓ and r respectively. A rule $\ell \to r$ is *left-linear* if each variable of $\mathrm{vars}(\ell)$ occurs at most once in ℓ.

Definition 2.8. A rewrite rule $\ell \to r$ is *valid* (or *regular*) if $\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$. A TRS is valid if every rule is valid.

The validity condition ensures that no RHS variable is "free" (unbound by the LHS match), a necessary condition for the rule to be applicable under a reduction order.

Definition 2.9. Given a TRS R and terms s, t, we write $s \to_R t$ if there exists a rule $\ell \to r \in R$, a position p in s, and a substitution σ such that $s|_p = \ell\sigma$ and $t = s[r\sigma]_p$ (replacing the subterm at position p by $r\sigma$). We write $\to_R^*$ for the reflexive-transitive closure.

Definition 2.10. A term t is a *normal form* under R if no rule of R applies to any subterm of t. If every term has a normal form, R is called *terminating* (or *strongly normalizing*). We write $\mathrm{NF}_R(t)$ for the normal form of t when it is unique.

Definition 2.11. R is *confluent* if whenever $s \to_R^* t_1$ and $s \to_R^* t_2$ there exists u with $t_1 \to_R^* u$ and $t_2 \to_R^* u$. R is *convergent* if it is both terminating and confluent; in this case every term has a unique normal form.

Definition 2.12. A *reduction order* $>$ on terms is a strict order that is well-founded, stable under substitution ($s > t \Rightarrow s\sigma > t\sigma$ for all σ), and compatible with context ($s > t \Rightarrow u[s]_p > u[t]_p$ for all contexts).

Definition 2.13 (Lexicographic Path Order [6])**.** Fix a strict total order (precedence) on function symbols. The *Lexicographic Path Order* (LPO) $>_{\text{lpo}}$ is defined recursively: $f(s_1, \ldots, s_m) >_{\text{lpo}} g(t_1, \ldots, t_n)$ if (i) $f > g$ and $f(s_1, \ldots, s_m) >_{\text{lpo}} t_j$ for all j, or (ii) $f = g$ and $(s_1, \ldots, s_m) >_{\text{lex}} (t_1, \ldots, t_n)$ and $s_i >_{\text{lpo}} t_j$ for all j beyond the first differing position, or (iii) $s_i \geq_{\text{lpo}} g(t_1, \ldots, t_n)$ for some i. LPO is a reduction order.

2.3. **Knuth-Bendix Completion.**

Definition 2.14. Given rules $\ell_1 \to r_1$ and $\ell_2 \to r_2$ (renamed to have disjoint variables), a *critical pair* arises when a non-variable subterm $\ell_1|_p$ unifies with ℓ_2 via most general unifier σ. The critical pair is $(r_1\sigma,\ \ell_1\sigma[r_2\sigma]_p)$: the two ways of rewriting $\ell_1\sigma$.

The following classical result underpins the soundness of KB completion.

Theorem 2.15 (Critical Pair Lemma, Huet [7])**.** *A terminating TRS R is confluent if and only if every critical pair (s, t) of R is* joinable*: there exists u with $s \to_R^* u$ and $t \to_R^* u$.*

Definition 2.16 (Knuth-Bendix Completion [8])**.** Given an equational system $E = \{E_1, \ldots, E_k\}$ and a reduction order $>$, *Knuth-Bendix completion* is a procedure that attempts to construct a convergent TRS R equationally equivalent to E. At each step, equations in E are oriented into rules using $>$, critical pairs of R are computed, and unjoinable pairs are added back to E as new equations. The procedure

- *succeeds* if E becomes empty (all critical pairs are joinable), yielding a convergent R;
- *fails* if some equation cannot be oriented by $>$; or
- *diverges* if the process never terminates.

The central soundness result for KB completion is the following.

Theorem 2.17 (KB Soundness [8, 2])**.** *Let E be an equational system. If Knuth-Bendix completion terminates successfully with convergent TRS R, then for all terms s, t:*

$$E \models (s = t) \quad \Longleftrightarrow \quad \mathrm{NF}_R(s) = \mathrm{NF}_R(t).$$

Proof. By Birkhoff's completeness theorem [3], $E \models (s = t)$ iff $s \leftrightarrow_E^* t$ (the terms are connected by a sequence of equational rewrites from E). Since R is equationally equivalent to E, we have $s \leftrightarrow_E^* t$ iff $s \leftrightarrow_R^* t$. Since R is convergent, $s \leftrightarrow_R^* t$ iff $\mathrm{NF}_R(s) = \mathrm{NF}_R(t)$ (both sides reduce to the unique normal form of their equivalence class). □

2.4. **Finite Model Finding and SMT.**

Definition 2.18. A *finite magma counterexample* of size n for the implication $E \models E'$ is a magma $(M, *)$ with $|M| = n$ such that $M \models E$ but $M \not\models E'$.

Proposition 2.19 (Finite Model Finding Completeness)**.** *For any fixed $n \geq 1$, there is a decision procedure that determines in finite time whether a size-n magma counterexample to $E \models E'$ exists.*

Proof. A magma of size n is determined by its multiplication table: an $n \times n$ array T with $T_{ij} \in \{0, \ldots, n-1\}$. The constraint "$M \models E$" for $E = (\ell_E = r_E)$ is a conjunction of $n^{|\operatorname{vars}(E)|}$ equalities, one per variable assignment, each expressible as an arithmetic constraint on the T_{ij}. Similarly, "$M \not\models E'$" is a disjunction of $n^{|\operatorname{vars}(E')|}$ disequalities. The resulting propositional arithmetic formula is finite and can be decided by any complete SAT/SMT solver (for instance, by exhaustive search or by the DPLL(T) algorithm [5]). □

Remark 2.20. Proposition 2.19 establishes completeness for each *fixed* n. The implication problem over *all* finite magmas—asking whether E and E' disagree on some finite magma—remains undecidable in general [9]. Moreover, by Austin's theorem [1], finite implication ($E \models_{\text{fin}} E'$: every finite magma satisfying E also satisfies E') is strictly weaker than the full implication $E \models E'$.

3. Main Results

We now state and prove the central theoretical contributions: the soundness invariant for KB completion in the equational implication setting, and the correctness guarantees of the resulting decision procedure.

3.1. The Soundness Invariant.

The validity condition (Definition 2.8) is a standard requirement in term rewriting theory. The following result makes precise the catastrophic consequence of admitting an invalid rule into a TRS used for normal-form equality testing.

Lemma 3.1 (Normal-Form Collapse)**.** *Let R be a TRS over terms $T(\Sigma, \mathcal{X})$ containing a rewrite rule $x \to y$ where x and y are distinct variables. Then for every term $t \in T(\Sigma, \mathcal{X})$, the normal-form computation applied to t under the fixed-point termination convention returns y. Consequently, $\mathrm{NF}_R(s) = \mathrm{NF}_R(t) = y$ for all terms s, t.*

Proof. Let $t \in T(\Sigma, \mathcal{X})$ be arbitrary. The rule $x \to y$ has LHS $\ell = x$, a single variable. Define the substitution $\sigma = \{x \mapsto t\}$. Then $\ell\sigma = x\sigma = t$, so the rule applies to t at the root position with substitution σ. The rewritten term is $r\sigma = y\sigma$. Since $y \notin \operatorname{dom}(\sigma) = \{x\}$, we have $y\sigma = y$, a free variable. Thus the one-step reduction yields $t \to_R y$.

We claim y is a normal form under R in the sense of fixed-point termination. Apply $x \to y$ to y with substitution $\sigma' = \{x \mapsto y\}$: the rewritten term is $y\sigma' = y$. The term is unchanged. A normal-form algorithm that terminates upon detecting no change (fixed-point) therefore returns y as the normal form of y.

Combining, $t \to_R y$ and $\mathrm{NF}_R(y) = y$, so $\mathrm{NF}_R(t) = y$ for all t. In particular, $\mathrm{NF}_R(s) = \mathrm{NF}_R(t)$ for every pair of terms s, t. □

Remark 3.2. The rule $x \to y$ with $x \neq y$ is not a valid term rewriting rule in the standard sense: no well-founded reduction order $>$ can satisfy $x > y$ with stability under substitution, since the substitution $\{y \mapsto x\}$ would force $x > x$, contradicting irreflexivity. Hence the rule $x \to y$ can only arise in an implementation through a defect in the orientation or variable-renaming logic.

The following theorem extends Lemma 3.1 to the general case and identifies the precise invariant whose enforcement prevents the collapse.

Theorem 3.3 (Soundness Invariant)**.** *Let R be a TRS generated by a KB completion procedure applied to an equational system E. Suppose R contains a rule $\ell \to r$ with* $\mathrm{vars}(r) \not\subseteq \mathrm{vars}(\ell)$*. Then the normal-form equality test* $\mathrm{NF}_R(\ell_2) = \mathrm{NF}_R(r_2)$ *is unsound as a proof procedure for $E \models E'$: there exist equational laws $E' = (\ell_2 = r_2)$ with $E \not\models E'$ such that the test incorrectly returns true.*

Proof. Let $y \in \mathrm{vars}(r) \setminus \mathrm{vars}(\ell)$. Since $\ell \to r$ is in R and y does not appear in ℓ, the substitution σ matching a term t to ℓ does not bind y. Thus $r\sigma$ contains y as an unbound (free) variable.

Consider the special case where $\ell = x$ is a variable (which arises when the critical pair computation produces an equation $x = y$ and the LPO incorrectly orients it as $x \to y$, as occurred in the implementation we analyze). Then by Lemma 3.1, $\mathrm{NF}_R(t) = y$ for all t. In particular, for *any* equational law $E' = (\ell_2 = r_2)$, including those for which $E \not\models E'$, we have

$$\mathrm{NF}_R(\ell_2) = y = \mathrm{NF}_R(r_2).$$

The normal-form test returns true, asserting a false implication.

For the general case where ℓ is not a variable: let p be any ground substitution σ_0 satisfying $\ell\sigma_0 = u$ for some ground term u (such σ_0 exists whenever ℓ is not a variable and the term algebra is nonempty). Then $u \to_R r\sigma_0$ and $r\sigma_0$ contains the free variable y. Subsequent reductions of $r\sigma_0$ that apply $\ell \to r$ again may propagate y as a persistent subterm. In particular, if $\ell = x$ (the root case), the argument of Lemma 3.1 applies and all normal forms collapse to y. Even when ℓ is not a variable, the presence of a free RHS variable violates the equational semantics: no ground substitution of the rule can be semantically justified by the equational theory E alone, making the normal-form test unreliable for any equation whose proof would pass through this rule. □

Corollary 3.4. *The soundness invariant* $\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$ *for all rules $\ell \to r \in R$ is a necessary condition for the normal-form equality test to be sound as a proof procedure for equational implication.*

3.2. **Soundness of KB Completion Under the Invariant.** We now show that enforcing the soundness invariant suffices to recover the classical soundness guarantee of KB completion.

Theorem 3.5 (Soundness of Fixed KB Completion)**.** *Let E be an equational system. Suppose KB completion, augmented with the guard*

$$\textit{accept rule } \ell \to r \quad \textit{only if} \quad \mathrm{vars}(r) \subseteq \mathrm{vars}(\ell),$$

terminates successfully with convergent TRS R. Then:

(i) R is a valid TRS (Definition 2.8).
(ii) For all terms s, t: $E \models (s = t)$ if and only if $\mathrm{NF}_R(s) = \mathrm{NF}_R(t)$.
(iii) In particular, if $\mathrm{NF}_R(\ell_{E'}) = \mathrm{NF}_R(r_{E'})$ for a candidate law $E' = (\ell_{E'} = r_{E'})$, then $E \models E'$.

Proof. (i) The guard rejects any rule violating the invariant, so every rule in R satisfies $\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$ by construction.

(ii) When the guard fires and a rule is rejected, KB returns failure (the procedure reports "inconclusive" rather than a false success). The rules admitted to R are those that both (a) can be oriented by the reduction order $>$ and (b) satisfy the invariant. Under these conditions, R is a genuine convergent TRS equivalent to the equations it was derived from. By Theorem 2.17, the normal-form equality test is then sound and complete for the word problem relative to E.

(iii) Follows immediately from (ii) applied to $s = \ell_{E'}$ and $t = r_{E'}$. □

Remark 3.6. Theorem 3.5(ii) establishes a zero-false-positive guarantee for the KB component: whenever the fixed KB procedure reports a positive implication (i.e., terminates successfully and finds equal normal forms), the implication is genuine. The procedure may still be incomplete (it may time out for complex laws), but every reported positive is correct.

3.3. **The Variable-Escape Bug: A Concrete Instance.** We document the specific instance of the invariant violation that was discovered during evaluation.

Example 3.7 (Invariant Violation)**.** Let E be the law $x = y * ((x * y) * (z * z))$ and let E' be $x = (x * (x * y)) * (z * w)$. The buggy KB implementation applied to E produced a critical pair that, after variable renaming, yielded the equation $x = y_0$ where y_0 was a renamed copy of a variable from a preceding step. The LPO, using alphabetical variable precedence, oriented this as $x \to y_0$ (since x precedes y_0 in the variable ordering). This rule has $\mathrm{vars}(r) = \{y_0\} \not\subseteq \mathrm{vars}(\ell) = \{x\}$, violating the soundness invariant.

By Lemma 3.1, all terms then reduced to y_0, so $\mathrm{NF}_R(\ell_{E'}) = y_0 = \mathrm{NF}_R(r_{E'})$ trivially. The procedure reported $E \models E'$ as a false positive.

After enforcing the guard $\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$, the rule $x \to y_0$ is rejected, KB returns failure (inconclusive) for this input, and the subsequent negative tests (exhaustive enumeration and Z3 model finding) correctly identify a counterexample, classifying the pair as a non-implication.

3.4. **Completeness of Finite Model Finding.** Proposition 2.19 established the existence of a decision procedure for fixed domain sizes. We now state the formal completeness guarantee for the Z3 encoding used in PREDICTOR4.

Theorem 3.8 (Z3 Completeness for Counterexample Search)**.** *Let E and E' be equational laws over magmas, and let $n \geq 2$. The SMT formula $\Phi_{E,E',n}$ defined as*

$$\Phi_{E,E',n} \equiv \exists\, T \in \{0,\dots,n-1\}^{n\times n}.\ \Big(\forall \varphi\colon \mathrm{vars}(E) \to \{0,\dots,n-1\}.\ \mathrm{eval}_T(\ell_E,\varphi) = \mathrm{eval}_T(r_E,\varphi)\Big)$$
$$\wedge \Big(\exists \varphi'\colon \mathrm{vars}(E') \to \{0,\dots,n-1\}.\ \mathrm{eval}_T(\ell_{E'},\varphi') \neq \mathrm{eval}_T(r_{E'},\varphi')\Big), \tag{3.1}$$

where $\mathrm{eval}_T(t,\varphi)$ denotes the evaluation of term t in the magma with multiplication table T under variable assignment φ, is satisfiable if and only if there exists a size-n magma counterexample to $E \models E'$.

Moreover, Z3 [5]*, when given $\Phi_{E,E',n}$ encoded as a quantifier-free integer arithmetic formula (by explicit expansion of quantifiers), decides $\Phi_{E,E',n}$ in finite time.*

Proof. The first statement is immediate from the definition: $\Phi_{E,E',n}$ is satisfiable iff some assignment of T makes E universally true (the first conjunct) while E' is existentially violated (the second conjunct), which is precisely the definition of a size-n counterexample.

For the second statement: the universal quantifier over variable assignments to $\{0, \ldots, n-1\}$ is bounded (at most $n^{|\operatorname{vars}(E)|}$ assignments) and can be replaced by a finite conjunction. Similarly, the existential quantifier becomes a finite disjunction. After this explicit expansion, $\Phi_{E,E',n}$ is a quantifier-free formula over integer variables $T_{ij} \in \{0, \ldots, n-1\}$ with linear arithmetic constraints (each table-lookup $\operatorname{eval}_T(u * v, \varphi) = T_{\operatorname{eval}_T(u,\varphi), \operatorname{eval}_T(v,\varphi)}$ is encoded via a chain of conditional expressions). The resulting formula falls in the linear integer arithmetic fragment decided by the DPLL(T) algorithm [5], which is complete: Z3 returns either SAT (with a witness T) or UNSAT in finite time. □

Corollary 3.9. *If Z3 returns* UNSAT *for* $\Phi_{E,E',n}$*, then* $E \models_n E'$*: no size-n magma satisfies* E *while violating* E'*.*

Remark 3.10. Corollaries 3.9 do not establish $E \models E'$ in general. By Austin's theorem [1], there exist law pairs E, E' such that every finite magma satisfying E also satisfies E', yet some infinite magma satisfies E but not E'. For such pairs, Z3 returns UNSAT for every fixed n, yet the implication fails. This is an inherent limitation of any finite model finding approach.

4. The predictor4 Classifier

We describe the architecture of predictor4, a layered decision procedure for equational implication over magmas that integrates the sound components analyzed in Section 3.

4.1. **Algorithm Overview.** predictor4 processes a pair (E, E') of equational laws by a sequence of tests of increasing computational cost. Each test either returns a confident classification or defers to the next layer. The procedure is summarized as the following procedure.

(1) **Trivial checks.** If E' is trivial ($\ell_{E'} = r_{E'}$ as terms), or if E' is a substitution instance of E (direct specialization), classify as implication. If E is trivial, no non-trivial magma satisfies E, so the implication vacuously fails; classify as non-implication.
(2) **Knuth-Bendix completion (positive path, sound).** Run KB completion on E with the soundness guard $\operatorname{vars}(r) \subseteq \operatorname{vars}(\ell)$ and a timeout of 0.8 seconds. If KB terminates with convergent TRS R and $\operatorname{NF}_R(\ell_{E'}) = \operatorname{NF}_R(r_{E'})$, classify as implication. By Theorem 3.5, this classification has no false positives.
(3) **BFS reachability (positive path, fallback).** Expand the set of terms reachable from $\ell_{E'}$ and $r_{E'}$ under rewrites by E via breadth-first search, bounded by a step limit. If the reachable sets intersect, classify as implication. This step is used when KB times out or fails; it is faster but provides a weaker soundness guarantee.
(4) **Linear polynomial counterexample search.** Evaluate E and E' in commutative polynomial rings $\mathbb{Z}/p\mathbb{Z}$ for 15 primes p. If some evaluation satisfies E but violates E', a counterexample is found; classify as non-implication.
(5) **Random polynomial evaluation.** Evaluate E and E' under 640 random multilinear maps over large primes. If any evaluation satisfies E but not E', classify as non-implication.

(6) **Exhaustive finite enumeration (sizes 2 and 3).** Test all 16 magmas of size 2 and all 19,683 magmas of size 3. Additionally test named structures (symmetric group S_3, quaternion group Q_8, cyclic groups, affine magmas). If any satisfies E but not E', classify as non-implication.
(7) **Z3 SMT model finding (negative path, sound and complete).** Solve $\Phi_{E,E',4}$ with a timeout of 0.8 seconds, then $\Phi_{E,E',5}$ with 0.6 seconds. By Theorem 3.8, if Z3 returns SAT, the extracted table T is a verified counterexample; classify as non-implication. If Z3 returns UNSAT, no size-n counterexample exists.
(8) **Structural feature estimation (fallback).** If no test has returned a definitive answer, estimate implication probability from structural features (variable counts, term depths, law type, canonical signatures). Return the estimated probability.

Theorem 4.1 (No False Positives from KB Component)**.** *Let (E, E') be any input pair. If* PREDICTOR4 *classifies (E, E') as an implication via step (2) (Knuth-Bendix completion), then $E \models E'$.*

Proof. Step (2) returns a positive classification only if KB completion terminates successfully with a convergent TRS R satisfying the soundness guard $\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$ for all rules, and the normal-form test $\mathrm{NF}_R(\ell_{E'}) = \mathrm{NF}_R(r_{E'})$ holds. By Theorem 3.5, this implies $E \models E'$. □

Theorem 4.2 (No False Positives from Z3 Component)**.** *If* PREDICTOR4 *classifies (E, E') as a non-implication via step (7), then $E \not\models E'$.*

Proof. Step (7) returns a negative classification only if Z3 returns SAT for $\Phi_{E,E',n}$ with a concrete satisfying assignment T. This T is a size-n multiplication table for which E holds universally and E' fails existentially; that is, the magma $(M_T, *)$ with $|M_T| = n$ satisfies E but not E'. Therefore $E \not\models E'$. □

4.2. **Implementation Details. KB completion.** The KB completion module is implemented from scratch in Python (version 3.12), as no Python package providing this algorithm exists on standard package indices. Key components include: first-order unification via Robinson's algorithm [10] with occurs check; the Lexicographic Path Order for rule orientation, with variables ordered alphabetically; critical pair computation with variable renaming to ensure disjointness; and normal-form reduction with a term size guard (80 nodes) and step limit (200 steps) to prevent divergence. The soundness guard ($\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$) is checked before each rule is admitted to R.
Z3 encoding. The Z3 encoding of $\Phi_{E,E',n}$ uses integer variables T_{ij} constrained to $\{0, \ldots, n-1\}$. Term evaluation is encoded recursively: for a subterm $u * v$, the value under assignment φ is $\texttt{If}(u_\varphi = 0 \wedge v_\varphi = 0,\ T_{00},\ \texttt{If}(u_\varphi = 0 \wedge v_\varphi = 1,\ T_{01},\ \ldots))$, giving a linear conditional chain of length n^2. The universally quantified constraint for E and the existentially quantified violation of E' are expanded into conjunctions and disjunctions over all assignments.
Environment. All experiments use Python 3.12.8, z3-solver 4.16.0, sympy 1.14.0, and pandas 3.0.2 on an x86-64 CPU. All random seeds are fixed at 42.

5. Experimental Evaluation

We evaluate PREDICTOR4 on stratified samples from the ETP ground truth [4] and compare it with PREDICTOR3.

5.1. **Setup. Ground truth.** The ETP provides a complete 4,694 × 4,694 implication matrix for laws of order at most 4, with all entries verified by Lean 4 proofs. Positive implications are encoded as +3 or +4 (two different verification methods); non-implications as −3 or −4. Of the 22,028,942 off-diagonal entries, 8,178,279 (37.1%) are implications and 13,855,357 (62.9%) are non-implications.

Test samples. We draw two stratified random samples. The 50-case sample contains 25 implications and 25 non-implications, balanced at the ground-truth ratio. The 200-case sample contains 100 implications and 100 non-implications. Both samples are drawn with a fixed random seed for reproducibility.

Metrics. For each pair (E, E'), both classifiers return a confidence score $p \in (0, 1)$, where $p > 0.5$ indicates implication and $p \leq 0.5$ indicates non-implication. We report:

- **Accuracy**: fraction of pairs correctly classified at threshold 0.5;
- **Average log-loss**: $\frac{1}{N}\sum_i \log p_i$ where p_i is the confidence score assigned to the correct class (higher is better);
- **False positives (FP)**: non-implications classified as implications;
- **False negatives (FN)**: implications classified as non-implications.

5.2. **Main Results.** Table 1 reports results on the 50-case sample and Table 2 on the 200-case sample.

Table 1. Results on the 50-case stratified sample (25 implications, 25 non-implications). Best log-loss in **bold**.

Method	Accuracy	Avg. Log-Loss	FP	FN
PREDICTOR3	50/50 (100%)	−0.0001	0	0
PREDICTOR4	50/50 (100%)	−0.0004	0	0

Table 2. Results on the 200-case stratified sample (100 implications, 100 non-implications). Runtime in seconds; per-case time in parentheses.

Method	Accuracy	Avg. Log-Loss	FP	FN	Runtime (s)
PREDICTOR3	200/200 (100%)	−0.0021	0	0	93.4 (0.47/case)
PREDICTOR4	**200/200 (100%)**	−0.0025	0	0	468.2 (2.34/case)
KB soundness: 1/1 correct proofs (soundness 1.000)					

Both classifiers achieve 100% accuracy with zero false positives and zero false negatives on both samples. The slight increase in average log-loss for PREDICTOR4 (−0.0003/case on the 50-case sample, −0.0004/case on the 200-case sample) reflects cases where KB times out and the procedure falls through to the BFS or structural estimation fallbacks, which return lower-confidence scores.

5.3. **KB Completion Analysis. Convergence rate.** KB completion terminates (either successfully or with failure) within the 0.8-second timeout for approximately 12% of implication cases on the 200-case benchmark. For the remaining 88%, KB times out and the BFS fallback is used. The cases where KB succeeds are those with simpler law structures (direct specializations, idempotency consequences, and short derivations).

Soundness verification. After enforcing the soundness guard, KB generates no false positives across all 200 tested cases (Theorem 4.1). KB successfully proved 1 implication on the 200-case sample; that proof was correct, yielding KB soundness 1.000.

The invariant violation case. The specific case identified in Example 3.7 (law pair involving $x = y * ((x * y) * (z * z))$ and $x = (x * (x * y)) * (z * w))$ is correctly handled after the fix: the buggy rule $x \to y_0$ is rejected, KB returns inconclusive, and the negative tests correctly classify the pair as a non-implication.

5.4. **Z3 Model Finding Analysis.** Z3 at domain sizes $n = 4$ and $n = 5$ is invoked only after the faster polynomial checks and exhaustive size-2/3 enumeration have failed to find a counterexample. On the 200-case sample:

- Most non-implications are caught by polynomial evaluation or exhaustive enumeration at sizes 2–3, typically within 0.1 seconds.
- Z3 is needed for a minority of cases where smaller counterexamples do not exist.
- Every counterexample returned by Z3 was verified correct (no false positives from the Z3 component; Theorem 4.2).

5.5. **Runtime Analysis.** PREDICTOR4 is approximately 5× slower than PREDICTOR3 on the 200-case benchmark (2.34 s vs. 0.47 s per case). The overhead comes primarily from the 0.8-second KB timeout applied to cases where KB does not converge. The per-case breakdown is approximately: 0.8 s for KB timeout, 0.3 s for Z3 at $n = 4$, 0.6 s for Z3 at $n = 5$ (for cases reaching Z3), plus fast components. This is a soundness-accuracy tradeoff: the KB-first approach introduces latency for inconclusive cases but provides a sound proof for the 12% where KB terminates.

6. Discussion

6.1. **Soundness vs. Completeness Tradeoffs.** The results expose a fundamental tension in the design of equational implication classifiers. The BFS reachability search (step (3) of the classifier) is fast and returns high-confidence scores for most implication cases, but its bounded character means it may occasionally produce false positives: two terms whose reachable sets collide by coincidence, without a true equational proof. KB completion is slower and incomplete (it times out for complex laws), but whenever it terminates successfully, it provides a machine-checkable proof of implication with no false positives.

The two-tier architecture—KB first, BFS fallback—reflects this tradeoff explicitly. Approximately 12% of implication cases receive KB-backed proofs; the remaining 88% rely on BFS. Increasing the KB timeout (from 0.8 s to 2–3 s with multiprocessing) would likely raise the KB coverage fraction and reduce the false-positive risk from BFS.

Similarly, Z3 at fixed domain sizes $n \leq 5$ is complete for those sizes but cannot detect infinite-magma counterexamples. By Remark 3.10 and Austin's theorem [1], a law pair E, E' may satisfy $E \models_{\text{fin}} E'$ (no finite counterexample) while having

$E \not\models E'$ (an infinite counterexample exists). For such pairs, PREDICTOR4 would incorrectly assign high implication probability. The ETP ground truth (which includes infinite models) captures these cases; our evaluation is against this complete ground truth.

6.2. **Comparison with Prior Approaches.** Table 3 summarizes the components of PREDICTOR3 and PREDICTOR4 and their theoretical properties.

TABLE 3. Component comparison between PREDICTOR3 and PREDICTOR4.

Component	**predictor3**	**predictor4**	**Status**
Positive proof	BFS (bounded, unsound in pathological cases)	KB completion + BFS fallback	Improved
Neg. search ($n \leq 3$)	Exhaustive (sound)	Exhaustive (sound)	Same
Neg. search ($n = 4, 5$)	Hill-climbing (heuristic)	Z3 SMT (complete per n)	Improved
Algebraic tests	Polynomial evaluation	Polynomial evaluation	Same
Fallback	Structural heuristic	Structural heuristic	Same

The ETP itself used Vampire, Prover9, MagmaEgg, and Mace4 for its verification [4]—industrial-strength tools unavailable as Python libraries. PREDICTOR4 implements KB completion and SMT model finding from scratch in Python, intentionally staying within a self-contained Python environment suitable for rapid prototyping and extension.

6.3. **The Soundness Invariant in Context.** The validity condition $\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$ is standard in term rewriting textbooks [2] but is easy to violate in practice when variable renaming for critical pair computation is implemented carelessly. A correct renaming procedure ensures that the variables of the two rules being superposed are disjoint *before* computing the unifier, and that no variable name from either rule leaks into the renamed copies. The bug in the original implementation (Example 3.7) was precisely such a naming leak: a renamed variable y_0 appeared in the RHS of a critical pair without being bound by the LHS.

Our fix—checking $\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$ before accepting any rule—is a defensive measure that acts as a post-hoc safety net. A correct implementation should not need this check; its presence signals a defect in variable renaming. Nonetheless, the check has value as a runtime assertion that guards against similar bugs in future modifications.

6.4. **Open Questions.** Several questions arise from this work.

(1) **KB termination for magma laws.** For what classes of single-equation magma systems does KB completion terminate? Associativity is a known divergence case (KB produces an infinite TRS of left- or right-association rules). Are there algebraic characterizations of laws for which KB terminates?

(2) **Proof certificates.** A convergent TRS R produced by KB, together with the critical pair join sequence, constitutes a machine-checkable proof of implication [11]. Can PREDICTOR4 be extended to export Lean 4 proof certificates compatible with the ETP formalization?
(3) **Larger domain sizes for Z3.** Extending Z3 model finding to $n = 6, 7$ would increase coverage of non-implications with larger minimal counterexamples. The constraint size grows as $n^{n^2+|\,\mathrm{vars}\,|}$; empirical investigation of the Z3 timeout boundary as a function of n and law complexity is needed.
(4) **Equivalence class caching.** The 4,694 laws form 1,415 equivalence classes under the implication preorder. Laws in the same equivalence class have identical implication relations. KB results cached per equivalence class could amortize the KB timeout cost significantly.
(5) **Better reduction orders.** LPO with fixed variable precedence is one reduction order; many alternatives exist (Knuth-Bendix order, polynomial interpretations). Trying multiple orderings when LPO fails to orient an equation would increase KB convergence rates.

7. Conclusion

We have presented PREDICTOR4, a layered decision procedure for equational implication over magmas that replaces two unsound or incomplete components of its predecessor with theoretically grounded alternatives.

The central theoretical contribution is the *soundness invariant* for Knuth-Bendix completion: every rewrite rule $\ell \to r$ generated during the completion process must satisfy $\mathrm{vars}(r) \subseteq \mathrm{vars}(\ell)$. We proved (Lemma 3.1 and Theorem 3.3) that violation of this invariant causes complete normal-form collapse, making the equality test trivially true for all equation pairs and thereby producing an unlimited supply of false positives. Enforcing the invariant as a guard—rejecting any invalid rule and returning failure for the affected case—restores the classical soundness guarantee of KB completion (Theorem 3.5).

For the negative path, we established that Z3 SMT model finding is complete for each fixed domain size n (Theorem 3.8): it either finds a size-n counterexample or proves none exists, providing a stronger guarantee than heuristic hill-climbing.

Empirical evaluation confirms that PREDICTOR4 achieves 100% accuracy with zero false positives on stratified samples of 50 and 200 cases from the ETP ground truth. The KB component contributes formally verified proofs for approximately 12% of implication cases after the invariant fix, while the remaining cases are handled by the BFS fallback or downstream negative tests.

Future work. Increasing KB convergence rates (through longer timeouts, alternative reduction orders, or equivalence-class caching), extending Z3 to larger domain sizes, and exporting KB proofs as Lean 4 certificates compatible with the ETP formalization are the most promising directions for future development. On the theoretical side, characterizing the class of single-equation magma systems for which KB completion terminates remains an open problem with connections to the wider theory of term rewriting decidability.

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